I. Заполнить массив Y значениями $f(x_n)$, $x_n = 100\cos n, n = \overline{0,N}$. Принять $N = 200$.

Сравнить эффективность реализации на основе оператора `for` и метода `For` класса `System.Threading.Tasks.Parallel`. Полагать:

1.

$$f(x) = \sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{x^2}{x^2+k^2+j^2}\cos\big((k+j)x\big) \approx$$

$$\approx \sum_{k=1}^{max(20,[20|x|])}\ \sum_{j=1}^{max(20,[20|x|])}\frac{x^2}{x^2+k^2+j^2}\cos\big((k+j)x\big)$$

2.

$$f(x) = \sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{x^2}{x^2+2k^2+j^2}\sin\big((k+2j)x\big) \approx$$

$$\approx \sum_{k=1}^{max(20,[20|x|])}\ \sum_{j=1}^{max(20,[20|x|])}\frac{x^2}{x^2+2k^2+j^2}\sin\big((k+2j)x\big)$$

3.

$$f(x) = \sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{x^2}{x^2+k^2+2j^2}\cos kx\sin jx \approx$$

$$\approx \sum_{k=1}^{max(20,[20|x|])}\ \sum_{j=1}^{max(20,[20|x|])}\frac{x^2}{x^2+k^2+2j^2}\cos kx\sin jx$$

4.

$$f(x) = \sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{x^2k}{x^2+k^3+j^2}\cos\big((k+j)x\big) \approx$$

$$\approx \sum_{k=1}^{max(20,[20|x|])}\ \sum_{j=1}^{max(20,[20|x|])}\frac{x^2k}{x^2+k^3+j^2}\cos\big((k+j)x\big)$$

5.

$$f(x) = \sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{x^2j}{x^2+k^2+j^3}\sin\big((k+2j)x\big) \approx$$

$$\approx \sum_{k=1}^{max(20,[20|x|])}\ \sum_{j=1}^{max(20,[20|x|])}\frac{x^2j}{x^2+k^2+j^3}\sin\big((k+2j)x\big)$$

6.

$$f(x) = \sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{x^2(k+j)}{x^2+k^3+j^3}\cos kx\sin jx \approx$$

$$\approx \sum_{k=1}^{max(20,[20|x|])}\ \sum_{j=1}^{max(20,[20|x|])}\frac{x^2(k+j)}{x^2+k^3+j^3}\cos kx\sin jx$$

7.

$$f(x)=\sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{(x^2+x)(k-j)}{x^2+k^3+j^3}\sin kx\cos jx\approx$$
$$\approx\sum_{k=1}^{max(20,[20|x|])}\sum_{j=1}^{max(20,[20|x|])}\frac{(x^2+x)(k-j)}{x^2+k^3+j^3}\sin kx\cos jx$$

8.

$$f(x)=\sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{(x^2-x)(k+2j)}{x^2+k^3+j^3}\cos\big((k+2j)x\big)\approx$$
$$\approx\sum_{k=1}^{max(20,[20|x|])}\sum_{j=1}^{max(20,[20|x|])}\frac{(x^2-x)(k+2j)}{x^2+k^3+j^3}\cos\big((k+2j)x\big)$$

9.

$$f(x)=\sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{x^2(k^2+j^2)}{x^2+k^4+j^4}\cos\big((k-j)x\big)\approx$$
$$\approx\sum_{k=1}^{max(20,[20|x|])}\sum_{j=1}^{max(20,[20|x|])}\frac{x^2(k^2+j^2)}{x^2+k^4+j^4}\cos\big((k-j)x\big)$$

10.

$$f(x)=\sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{x^2(k^2+j)}{x^2+k^4+j^3}\sin\big((2k-j)x\big)\approx$$
$$\approx\sum_{k=1}^{max(20,[20|x|])}\sum_{j=1}^{max(20,[20|x|])}\frac{x^2(k^2+j)}{x^2+k^4+j^3}\sin\big((2k-j)x\big)$$

11.

$$f(x)=\sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{x^2(k+2j^2)}{x^2+k^3+j^4}\cos\big((k+2j)x\big)\approx$$
$$\approx\sum_{k=1}^{max(20,[20|x|])}\sum_{j=1}^{max(20,[20|x|])}\frac{x^2(k+2j^2)}{x^2+k^3+j^4}\cos\big((k+2j)x\big)$$